

ER Diagram — Research University System

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erDiagram
    USERS ||--o| STUDENTS : "is"
    USERS ||--o| EMPLOYEES : "is"
    EMPLOYEES ||--o| TEACHERS : "is"
    EMPLOYEES ||--o| MANAGERS : "is"
    EMPLOYEES ||--o| ADMINS : "is"

    STUDENTS ||--|| TRANSCRIPTS : "has"
    STUDENTS ||--o{ REGISTRATION_REQUESTS : "creates"
    COURSES ||--o{ REGISTRATION_REQUESTS : "requested_for"
    MANAGERS ||--o{ REGISTRATION_REQUESTS : "processes"

    STUDENTS ||--o{ ENROLLMENTS : "has"
    COURSES ||--o{ ENROLLMENTS : "has"

    COURSES ||--o{ LESSONS : "contains"
    COURSES ||--o{ MARKS : "has"
    STUDENTS ||--o{ MARKS : "receives"
    TRANSCRIPTS ||--o{ MARKS : "includes"

    COURSES ||--o{ COURSE_TEACHERS : "has"
    TEACHERS ||--o{ COURSE_TEACHERS : "teaches"

    STUDENTS ||--o{ TEACHER_RATINGS : "gives"
    TEACHERS ||--o{ TEACHER_RATINGS : "receives"

    USERS ||--o| RESEARCH_PROFILES : "may_have"
    RESEARCH_PROFILES ||--o{ RESEARCH_PAPER_AUTHORS : "writes"
    RESEARCH_PAPERS ||--o{ RESEARCH_PAPER_AUTHORS : "has_author"
    RESEARCH_PROFILES ||--o{ RESEARCH_PROJECT_PARTICIPANTS : "joins"
    RESEARCH_PROJECTS ||--o{ RESEARCH_PROJECT_PARTICIPANTS :
"has_participant"
    RESEARCH_PROJECTS ||--o{ RESEARCH_PROJECT_PAPERS : "publishes"
    RESEARCH_PAPERS ||--o{ RESEARCH_PROJECT_PAPERS : "belongs_to"

    EMPLOYEES ||--o{ MESSAGES : "sends"
    EMPLOYEES ||--o{ MESSAGES : "receives"
    MANAGERS ||--o{ NEWS : "creates"
    MANAGERS ||--o{ REPORTS : "generates"
    USERS ||--o{ USER_ACTION_LOGS : "performs"

    USERS {
        string id PK
        string username UK
        string password_hash
        string full_name
        string email UK
```

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}

STUDENTS {
    string user_id PK, FK
    int year_of_study
    double gpa
    int total_credits
}

EMPLOYEES {
    string user_id PK, FK
    double salary
    date hire_date
}

TEACHERS {
    string employee_id PK, FK
    string title
}

MANAGERS {
    string employee_id PK, FK
    string manager_type
}

ADMINS {
    string employee_id PK, FK
}

COURSES {
    string code PK
    string name
    int credits
}

LESSONS {
    int id PK
    string course_code FK
    string topic
    datetime date_time
    string lesson_type
}

REGISTRATION_REQUESTS {
    int id PK
    string student_id FK
    string course_code FK
    string processed_by_manager_id FK
    datetime created_at
    string status
}

```

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ENROLLMENTS {
    string student_id PK, FK
    string course_code PK, FK
    datetime enrolled_at
}

TRANSCRIPTS {
    int id PK
    string student_id UK, FK
}

MARKS {
    int id PK
    string student_id FK
    string course_code FK
    int transcript_id FK
    double first_attestation
    double second_attestation
    double final_exam
}

COURSE_TEACHERS {
    string course_code PK, FK
    string teacher_id PK, FK
}

TEACHER_RATINGS {
    int id PK
    string student_id FK
    string teacher_id FK
    int rating
    string comment
    datetime created_at
}

RESEARCH_PROFILES {
    int id PK
    string user_id UK, FK
    int h_index
}

RESEARCH_PAPERS {
    int id PK
    string title
    int pages
    date publication_date
    string doi UK
    int citations
}

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RESEARCH_PAPER_AUTHORS {
    int paper_id PK, FK
    int research_profile_id PK, FK
    int author_order
}

RESEARCH_PROJECTS {
    int id PK
    string topic
}

RESEARCH_PROJECT_PARTICIPANTS {
    int project_id PK, FK
    int research_profile_id PK, FK
}

RESEARCH_PROJECT_PAPERS {
    int project_id PK, FK
    int paper_id PK, FK
}

MESSAGES {
    int id PK
    string text
    datetime sent_at
    string sender_employee_id FK
    string receiver_employee_id FK
}

NEWS {
    int id PK
    string title
    string content
    datetime created_at
    string manager_id FK
}

REPORTS {
    int id PK
    string title
    datetime generated_at
    string manager_id FK
}

USER_ACTION_LOGS {
    int id PK
    string action
    datetime created_at
    string user_id FK
}

```

Notes

This ER diagram is based on the UML class diagram. Methods from UML are not included because ER diagrams describe data structure, not behavior.

Enums are represented as fields, not separate entities:

- `RegistrationStatus` → `REGISTRATION_REQUESTS.status`
- `LessonType` → `LESSONS.lesson_type`
- `TeacherTitle` → `TEACHERS.title`
- `ManagerType` → `MANAGERS.manager_type`

The `Researcher` interface from UML is represented by the `RESEARCH_PROFILES` table. A user becomes a researcher if they have a related research profile.